ELEC2302/9302 Signals and Systems — Final exam formula sheet

Complex exponentials

$$e^{j\theta} = \cos\theta + j\sin\theta, \qquad z = x + jy = r\,e^{j\theta}, \qquad r = |z| = \sqrt{x^2+y^2}, \qquad \theta = \arg z.$$

Inner product, energy and power

$$\langle u, y\rangle := \int_{-\infty}^{\infty} u(t)\,\overline{y}(t)\,\mathrm{d}t, \qquad E(u) = \int_{-\infty}^{\infty} |u(t)|^2\,\mathrm{d}t, \qquad P(u) = \lim_{W\to\infty} \frac{1}{2W}\int_{-W}^{W} |u(t)|^2\,\mathrm{d}t, \qquad u_{\mathrm{RMS}} = \sqrt{P(u)}.$$

If u is periodic with period T: $\langle u, y\rangle_P = \dfrac{1}{T}\displaystyle\int_0^T u(t)\,\overline{y}(t)\,\mathrm{d}t, \qquad P(u) = \langle u, u\rangle_P = \dfrac{1}{T}\displaystyle\int_0^T |u(t)|^2\,\mathrm{d}t.$

Standard signals and the Dirac delta

$$H(t) = \int_{-\infty}^{t} \delta(\varsigma)\,\mathrm{d}\varsigma = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} \qquad p_\tau(t) = \begin{cases} 1, & |t| \leq \tau/2 \\ 0, & \text{otherwise} \end{cases} = H\left(t+\tfrac{\tau}{2}\right) - H\left(t-\tfrac{\tau}{2}\right), \qquad \frac{\mathrm{d}}{\mathrm{d}t}H(t) = \delta(t).$$

$$\int_{-\infty}^{\infty} \delta(t-\tau)\,y(t)\,\mathrm{d}t = y(\tau), \qquad \delta(\alpha t) = \frac{1}{|\alpha|}\,\delta(t).$$

Convolution

$$(h * u)(t) := \int_{-\infty}^{\infty} h(t-\tau)\,u(\tau)\,\mathrm{d}\tau = (u * h)(t), \qquad (u * \delta)(t) = u(t), \qquad \big(u * \delta(t-T)\big) = u(t-T),$$

$$h_1 * (h_2 * h_3) = (h_1 * h_2) * h_3.$$

Impedances

$$Z_R = R, \qquad Z_L = j\omega L, \qquad Z_C = \frac{1}{j\omega C}.$$

Fourier series (u T-periodic)

Exponential form:

$$\phi_k(t) = e^{jk\omega_0 t}, \qquad c_k = \langle u, \phi_k\rangle_P = \frac{1}{T}\int_0^T u(t)\,e^{-jk\omega_0 t}\,\mathrm{d}t, \qquad u(t) = \sum_{k=-\infty}^{\infty} c_k\,e^{jk\omega_0 t}, \qquad c_0 = \frac{1}{T}\int_0^T u(t)\,\mathrm{d}t.$$

Sine–cosine form (u real): $u(t) = a_0 + \displaystyle\sum_{k=1}^{\infty}\big(a_k\cos(k\omega_0 t) + b_k\sin(k\omega_0 t)\big)$, where

$$a_0 = c_0, \quad a_k = 2\,\mathrm{Re}(c_k) = \frac{2}{T}\int_0^T u(t)\cos(k\omega_0 t)\,\mathrm{d}t, \quad b_k = -2\,\mathrm{Im}(c_k) = \frac{2}{T}\int_0^T u(t)\sin(k\omega_0 t)\,\mathrm{d}t, \quad c_k = \tfrac{1}{2}\left(a_k - jb_k\right).$$

u real	$c_{-k} = \overline{c_k}$	linearity	$\alpha u_1 + \beta u_2$ has coefficients $\alpha c_{1,k} + \beta c_{2,k}$
u even	$b_k = 0$	time shift	$u(t-t_0)$ has coefficients $e^{-jk\omega_0 t_0}\,c_k$
u odd	$a_0 = a_k = 0$	Parseval	$P(u) = \dfrac{1}{T}\displaystyle\int_0^T |u(t)|^2\,\mathrm{d}t = \sum_{k=-\infty}^{\infty} |c_k|^2$

Fourier series of standard waveforms

$\mathrm{sinc}(x) = \dfrac{\sin(\pi x)}{\pi x}$.

Signal	Coefficients
$\cos(\omega_0 t)$	$c_1 = c_{-1} = \frac{1}{2}$, all others zero
$\sin(\omega_0 t)$	$c_1 = -\frac{j}{2}$, $c_{-1} = \frac{j}{2}$, all others zero
Square wave, odd: $+A$ for $0 < t < T/2$, $-A$ for $-T/2 < t < 0$	$c_k = -\dfrac{2jA}{\pi k}$ (k odd), $c_k = 0$ (k even)
Pulse train: $A\,p_\tau(t)$ repeated with period T, duty cycle $d = \tau/T$	$c_k = A\,d\,\mathrm{sinc}(kd)$
Triangle wave: $A\,(1 - 4\lvert t\rvert/T)$ for $\lvert t\rvert \leq T/2$	$c_k = \dfrac{4A}{\pi^2 k^2}$ (k odd), $c_k = 0$ (k even)
Rectified sine: $A\,\lvert\sin(\omega_m t)\rvert$, period $T = \pi/\omega_m$, so $\omega_0 = 2\omega_m$	$c_k = \dfrac{2A}{\pi\,(1-4k^2)}$

Fourier transform

$$U(j\omega) = \int_{-\infty}^{\infty} u(t)\, e^{-j\omega t}\, \mathrm{d}t, \qquad u(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} U(j\omega)\, e^{j\omega t}\, \mathrm{d}\omega, \qquad u \text{ periodic: } U(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} c_k\, \delta(\omega - k\omega_0).$$

$u(t)$	$U(j\omega)$	$u(t)$	$U(j\omega)$
$\delta(t)$	1	$p_\tau(t)$	$\tau\,\mathrm{sinc}\left(\frac{\omega\tau}{2\pi}\right)$
$\delta(t-t_0)$	$e^{-j\omega t_0}$		
1	$2\pi\delta(\omega)$	$1-\frac{\lvert t\rvert}{\tau}$ for $\lvert t\rvert<\tau$	$\tau\,\mathrm{sinc}^2\left(\frac{\omega\tau}{2\pi}\right)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$	$e^{-at}H(t),\ a>0$	$\frac{1}{a+j\omega}$
$\cos(\omega_0 t)$	$\pi\delta(\omega-\omega_0)+\pi\delta(\omega+\omega_0)$	$t\,e^{-at}H(t)$	$\frac{1}{(a+j\omega)^2}$
$\sin(\omega_0 t)$	$-j\pi\delta(\omega-\omega_0)+j\pi\delta(\omega+\omega_0)$	$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2+\omega^2}$
$H(t)$	$\pi\delta(\omega)+\frac{1}{j\omega}$	$e^{-at}\cos(\omega_0 t)H(t)$	$\frac{a+j\omega}{(a+j\omega)^2+\omega_0^2}$
$\mathrm{sgn}(t)$	$\frac{2}{j\omega}$	$e^{-at}\sin(\omega_0 t)H(t)$	$\frac{\omega_0}{(a+j\omega)^2+\omega_0^2}$

Properties of the Fourier transform

	$u(t)$	$U(j\omega)$		$u(t)$	$U(j\omega)$
linearity	$\alpha u_1(t)+\beta u_2(t)$	$\alpha U_1(j\omega)+\beta U_2(j\omega)$	differentiation	$\frac{\mathrm{d}}{\mathrm{d}t}u(t)$	$j\omega\, U(j\omega)$
time shift	$u(t-t_0)$	$e^{-j\omega t_0}\, U(j\omega)$	integration	$\int_{-\infty}^{t} u(\varsigma)\,\mathrm{d}\varsigma$	$\frac{U(j\omega)}{j\omega}+\pi U(0)\delta(\omega)$
frequency shift	$e^{j\omega_0 t}u(t)$	$U\big(j(\omega-\omega_0)\big)$	multiply by t	$t\,u(t)$	$j\frac{\mathrm{d}}{\mathrm{d}\omega}U(j\omega)$
modulation	$u(t)\cos(\omega_0 t)$	$\frac{1}{2}U\big(j(\omega-\omega_0)\big)+\frac{1}{2}U\big(j(\omega+\omega_0)\big)$	convolution	$(h*u)(t)$	$H(j\omega)\,U(j\omega)$
time scaling	$u(at)$	$\frac{1}{\lvert a\rvert}U\left(\frac{j\omega}{a}\right)$	multiplication	$u(t)\,y(t)$	$\frac{1}{2\pi}(U*Y)(j\omega)$
time reversal	$u(-t)$	$U(-j\omega)$	duality	$U(jt)$	$2\pi\, u(-\omega)$

Parseval's theorem:

$$\int_{-\infty}^{\infty} u(t)\,\overline{y}(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} U(j\omega)\,\overline{Y}(j\omega)\,\mathrm{d}\omega, \qquad E(u) = \int_{-\infty}^{\infty} |u(t)|^2\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |U(j\omega)|^2\,\mathrm{d}\omega.$$

Second order systems

$$G(j\omega) = \frac{K\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega)+\omega_n^2}, \qquad G(0) = K,$$

where $\omega_n > 0$ is the natural frequency and ζ the damping ratio.

$\zeta > 1$	overdamped	distinct real poles
$\zeta = 1$	critically damped	repeated pole $-\omega_n$
$\zeta < 1$	underdamped	$-\zeta\omega_n \pm j\omega_d$, $\omega_d = \omega_n\sqrt{1-\zeta^2}$ the damped natural frequency

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}, \qquad |G(j\omega_r)| = \frac{|K|}{2\zeta\sqrt{1-\zeta^2}}, \qquad \zeta < \frac{1}{\sqrt{2}}.$$

Stability

Exponentially stable: there exist $M>0$ and $a>0$ such that $|h(t)| \le Me^{-at}$ for all $t \ge 0$.
Marginally stable: there exists $M>0$ such that $|h(t)| \le M$ for all $t \ge 0$, and $h(t) \not\to 0$ as $t \to \infty$.
BIBO stable: $\sup_{t\ge 0}|u(t)| < \infty$ implies $\sup_{t\ge 0}|y(t)| < \infty$ for every input u.

Laplace transform

$$U(s) = \mathcal{L}\{u\}(s) = \int_{0^-}^{\infty} u(t)\, e^{-st}\, \mathrm{d}t, \qquad Y(s) = G(s)\, U(s), \qquad G(s) = \mathcal{L}\{h\}(s).$$

$u(t)$	$U(s)$	$u(t)$	$U(s)$
$\delta(t)$	1	$e^{-at}H(t)$	$\dfrac{1}{s+a}$
$\delta(t-t_0),\ t_0 \geq 0$	e^{-st_0}	$t\, e^{-at}H(t)$	$\dfrac{1}{(s+a)^2}$
$H(t)$	$\dfrac{1}{s}$	$t^n e^{-at}H(t)$	$\dfrac{n!}{(s+a)^{n+1}}$
$t\, H(t)$	$\dfrac{1}{s^2}$	$e^{-at}\cos(\omega_0 t)H(t)$	$\dfrac{s+a}{(s+a)^2+\omega_0^2}$
$t^n H(t)$	$\dfrac{n!}{s^{n+1}}$	$e^{-at}\sin(\omega_0 t)H(t)$	$\dfrac{\omega_0}{(s+a)^2+\omega_0^2}$
$\cos(\omega_0 t)H(t)$	$\dfrac{s}{s^2+\omega_0^2}$	$\sin(\omega_0 t)H(t)$	$\dfrac{\omega_0}{s^2+\omega_0^2}$

Properties of the Laplace transform

	$u(t)$	$U(s)$		$u(t)$	$U(s)$
linearity	$\alpha u_1(t) + \beta u_2(t)$	$\alpha U_1(s) + \beta U_2(s)$	differentiation	$\dfrac{\mathrm{d}}{\mathrm{d}t}u(t)$	$s\,U(s) - u(0^-)$
time shift	$u(t-t_0)H(t-t_0),\ t_0 \geq 0$	$e^{-st_0}\, U(s)$	second derivative	$\dfrac{\mathrm{d}^2}{\mathrm{d}t^2}u(t)$	$s^2U(s) - s\,u(0^-) - \dot{u}(0^-)$
s-shift	$e^{-at}u(t)$	$U(s+a)$	integration	$\displaystyle\int_{0^-}^{t} u(\varsigma)\,\mathrm{d}\varsigma$	$\dfrac{U(s)}{s}$
time scaling	$u(at),\ a>0$	$\dfrac{1}{a}\,U\!\left(\dfrac{s}{a}\right)$	convolution	$(h*u)(t)$	$H(s)\,U(s)$
multiply by t	$t\,u(t)$	$-\dfrac{\mathrm{d}}{\mathrm{d}s}U(s)$	modulation	$u(t)\cos(\omega_0 t)$	$\frac{1}{2}U(s-j\omega_0) + \frac{1}{2}U(s+j\omega_0)$

Initial value theorem: $u(0^+) = \lim\limits_{s\to\infty} s\,U(s)$, **Final value theorem:** $\lim\limits_{t\to\infty} u(t) = \lim\limits_{s\to 0} s\,U(s)$.

Butterworth filters

$$|G(j\omega)|^2 = \frac{1}{1+(\omega/\omega_c)^{2n}}, \qquad n=1:\ \frac{\omega_c}{s+\omega_c}, \qquad n=2:\ \frac{\omega_c^2}{s^2+\sqrt{2}\,\omega_c s+\omega_c^2}, \qquad n=3:\ \frac{\omega_c^3}{(s+\omega_c)(s^2+\omega_c s+\omega_c^2)}.$$

$$n \text{ even: } G(s) = \prod_{k=1}^{n/2} \frac{\omega_c^2}{s^2+2\zeta_k\omega_c s+\omega_c^2}, \qquad n \text{ odd: } G(s) = \frac{\omega_c}{s+\omega_c}\prod_{k=1}^{(n-1)/2} \frac{\omega_c^2}{s^2+2\zeta_k\omega_c s+\omega_c^2}, \qquad \zeta_k = \sin\!\left(\frac{(2k-1)\pi}{2n}\right).$$

Modulation

AM	$y(t) = u(t)\cos(\omega_c t), \qquad Y(j\omega) = \frac{1}{2}U\big(j(\omega-\omega_c)\big) + \frac{1}{2}U\big(j(\omega+\omega_c)\big)$
Demodulation	$z(t) = y(t)\cos(\omega_c t) = \frac{1}{2}u(t) + \frac{1}{2}u(t)\cos(2\omega_c t)$
FM	$y(t) = \cos\!\Big(\omega_c t + k\displaystyle\int_{-\infty}^{t} u(\tau)\,\mathrm{d}\tau\Big)$
PM	$y(t) = \cos\big(\omega_c t + k\,u(t)\big)$

First and second order systems

First order, $G(s) = \dfrac{K}{\tau s+1}$: the step response reaches 63% of its final value at $t=\tau$, rise time (10% to 90%) $\approx 2.2\tau$, settling time (to within 5%) $\approx 3\tau$.

Second order, $G(s) = \dfrac{K\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$, $0<\zeta<1$: settling time (to within 5%) $\approx \dfrac{3}{\zeta\omega_n}$.

ζ	0.1	0.15	0.2	0.25	0.3	0.4	0.5	0.6	0.707
overshoot M_p	73%	62%	53%	44%	37%	25%	16%	9.5%	4.3%
resonant peak $\lvert G(j\omega_r)\rvert/\lvert K\rvert$	5.03	3.37	2.55	2.07	1.75	1.36	1.15	1.04	1.00
resonant peak, dB above DC	14.0	10.6	8.1	6.3	4.8	2.7	1.2	0.4	0